more parameters can definitely have wrong divisions between regions. For instance, imagine increasing the number of data points in Fig. 26-3, revealing a small overlap between the target and nontarget groups. It would be possible to move the threshold line between the groups to trade the number of false-positives against the number of false-negatives. That is, the diagonal line would be moved toward the top-right, or the bottom-left. However, it would be wrong to rotate the line, because it would increase *both* types of errors.

As suggested by these examples, the conventional approach to target detection (sometimes called pattern recognition) is a two step process. The first step is called **feature extraction**. This uses algorithms to reduce the raw data to a few parameters, such as diameter, brightness, edge sharpness, etc. These parameters are often called **features** or **classifiers**. Feature extraction is needed to reduce the amount of data. For example, a medical x-ray image may contain more than a million pixels. The goal of feature extraction is to distill the information into a more concentrated and manageable form. This type of algorithm development is more of an art than a science. It takes a great deal of experience and skill to look at a problem and say: "*These are the classifiers that best capture the information.*" Trial-and-error plays a significant role.

In the second step, an evaluation is made of the classifiers to determine if the target is present or not. In other words, some method is used to divide the parameter space into a region that corresponds to the targets, and a region that corresponds to the nontargets. This is quite straightforward for one and two-parameter spaces; the known data points are plotted on a graph (such as Fig. 26-3), and the regions separated by eye. The division is then written into a computer program as an equation, or some other way of defining one region from another. In principle, this same technique can be applied to a three-dimensional parameter space. The problem is, three-dimensional graphs are very difficult for humans to understand and visualize (such as Fig. 26-4). Caution: Don't try this in hyperspace; your brain will explode!

In short, we need a machine that can carry out a multi-parameter space division, according to examples of target and nontarget signals. This ideal target detection system is remarkably close to the main topic of this chapter, the *neural network*.

Neural Network Architecture

Humans and other animals process information with *neural networks*. These are formed from *trillions* of **neurons** (nerve cells) exchanging brief electrical pulses called **action potentials**. Computer algorithms that mimic these biological structures are formally called **artificial neural networks** to distinguish them from the squishy things inside of animals. However, most scientists and engineers are not this formal and use the term *neural network* to include both biological and nonbiological systems.